Study Plan v2 - Mohammed Mohsen Mohammed Al-Sakkaf (A23CS4026)

SCSEH01 Bachelor in Software Engineering with Honours, UTM - intake 2024/2025 - **as of 2026-10-14** - current semester 202620271 (SEM 5). This is a personal plan, not an official document. Tags: **P** = proven in a document, **C** = calculated, **A** = assumption.

Legend (status is always written as text; colour is only a helper): Passed | Registered (slip) | To add (planned, unconfirmed) | Repeat needed | Not yet taken | Elective slot TBD | Failed - superseded by repeat **Categories:** T faculty core | E programme elective | U university (U-FE = free-elective slot, counted inside U) | B English/languages | K co-curriculum | B* = outside the 128 (conflict K1). **Sources:** [AUD] Graduation-Audit.pdf; [RES] 2025-2026-Sem-2-Exam-Results.pdf; [SLIP] 2026-2027-Sem-1-Course-Registration.pdf; [CCS] Course-Completion-Status.pdf; [STR] SCSEH-Course-Structure-Intake-2024-2025-Rev-2024-11-05.pdf; [CUR] SCSEH-Curriculum-Course-List.pdf; [CAL] Academic-Calendar-2026-2027-Corrected-2026-05-14.pdf; [FL] Foreign-Languages-Timetable-2026-2027-Sem-1.pdf; [OLD] Mohammed-SCSEH-Study-Plan-2026-10-12.pdf; [MOH] Mohammed's own statements/corrections; [OMAR] Mohammed_SCSEH_Plan_review_2026-10-13.md

Credits by category (exactly 128) - C

Cat	Meaning	Required	Earned	Registered (slip)	To add (SEM 5)	Planned SEM 6-9	Remaining after plan	Still needed today
T	Faculty core (Wajib/Teras Fakulti)	82	49	0	0	33	0	33
E	Programme electives (Elektif)	18	0	9	0	9	0	18
U	University (Umum Universiti) incl. 4 Free Electives = 12	20	4	3	0	13	0	16
B	English / languages (Bahasa Inggeris)	6	0	0	6	0	0	6
K	Co-curriculum (Kokurikulum)	2	2	0	0	0	0	0
Total		**128**	**55**	**12**	**6**	**55**	**0**	**73**

U = 8 general (ULRS1032, UHLM1012, ULRS1022, ULRS3032) + 12 free electives (FE I to IV, 3 each; SFMX1063 = FE II). Earned 55 + UHLB1112 2 = official KD 57 (P). Audit 'Jumlah kredit dapat' also counts registered courses: T49 E9 U7 B2 K2. **UHLB1112 is outside these totals: see K1.**

KD projection per semester (if every course is passed) - C

Point	Official KD	Toward 128	Tag	Gate
Now - after 2025/26-II (official)	57	55	P [RES]	-
End SEM 5 - registered 12 only	69	67	C	< 80: SCSE3242 not possible in SEM 6
End SEM 5 - registered 12 + to add 6	75	73	C (A3)	< 80: SCSE3242 still not possible in SEM 6
End SEM 6 (+17)	92	90	C (A4)	>= 80: SCSE3242 OK in SEM 7; 92 IT gate reached (official KD)
End SEM 7 (+13)	105	103	C	
End SEM 8 (+13)	118	116	C	>= 92 + CGPA >= 2.0: Industrial Training OK in SEM 9
End SEM 9 (+12)	130	128	C	128 graduation credits reached (official KD 130 incl. UHLB1112, see K1)

What changed vs the old plan (2026-10-12)

- SEM 5 load: old block said 19 (counted a blank 'Free Elective II' on top of SFMX1063, while UHLB2122 sat in the Y2S2 block). Now: 12 registered + 6 to add (UHLB2122, UHLB3132, UHLF1112) = 18.
- SFMX1063 Information Technology in Business = category U (audit), filling the FE II slot. No second FE II. FE I (never taken in Y1S1) is now an explicit open slot, placed in SEM 7.
- Repeats SCSE2233 Software Design and Architecture and SCSP3213 Business Intelligence: old plan had no future slot. Now in SEM 6 (Sem II, the semester where [STR] offers them).
- ULRS1022 Philosophy and Current Issues (never taken) now placed in SEM 7.
- SCSE3242 Software Engineering Project I moved from SEM 6 to SEM 7: KD can reach at most 75 by the start of SEM 6 (<80), and 92 by SEM 7.
- SCSE3213 Software Quality and Testing moved from SEM 6 to SEM 8 (needs SCSE2233 passed first). SCSE4214 Software Engineering Project II stays in SEM 8. Industrial Training SCSE4108/SCSE4114 moves to SEM 9.
- 'SCSRxxx3 Program Elective' x2 replaced by the official Y4S2 options: SCSR4453, SCSR4973, SECB3133, SECB3203, PRISMS.
- UHLB1112 English Communication Skills no longer inside a semester total (old Y1S1 = 19). [STR] Y1S1 = 17; UHLB1112 is shown separately (conflict K1).
- Every row now has a text status, category, prerequisite check, source and evidence tag (P/C/A). Added category totals, KD projection and CGPA section.

CGPA

Item	Value	Tag
Current CGPA (PNGK), JMN 155.66, KK 61	2.55 (155.66 / 61 = 2.5518)	P / C
Ceiling formula this semester (h = new graded credits, all A)	(155.66 + 4h) / (61 + h)	C
h = 12 (12 registered): all A / all B / all C	2.79 / 2.63 / 2.46	C
h = 18 (12 + 6 to add): all A / all B / all C	2.88 / 2.65 / 2.43	C (A3)
Repeats SCSE2233 + SCSP3213 in SEM 6: if each D (3 cr x 1.00) is replaced, e.g. by B, from today's base	(155.66 - 6 + 18) / 61 = 2.75	A (A5)
Repeat replacement is an **assumption.** It is supported because JMN/KK match exactly when the first D attempts of SCSR2213 and SCST1223 are left out. Industrial Training needs CGPA ≥ 2.0: today 2.55 (P), so the margin is 0.55. SFMX1063 (NFE) and the UHLB/UHLF adds are assumed to be graded.		

Semester-by-semester plan (one block per semester)

Y1 S1 - 2024/2025 Sem I (202420251) - completed

Code	Course (English name)	Cr	Cat	Status	Grade / section	Prerequisite	Prereq met?	Evidence	Source	Note
SCSE1013	Fundamental Programming Concepts	3	T	Passed (C)	C	-	-	P	AUD	
SCST1123	Mathematics for Software Engineer	3	T	Passed (D+)	D+	-	-	P	AUD	
SCSR1013	Digital Logic	3	T	Passed (C+)	C+	-	-	P	AUD	
SCST1143	Database Engineering	3	T	Passed (C+)	C+	-	-	P	AUD	
ULRS1032	Integrity and Anti-Corruption Course	2	U	Passed (A+)	A+	-	-	P	AUD	
UHLB1112	English Communication Skills	2 (0 here)	B*	Passed (HL (status HW))	HL (status HW)	-	-	P	AUD/STR	Audit lists it under B and it is inside KD 57; [STR] makes it an extra course for low English scores. Kept OUTSIDE the 128 here (conflict K1)
FE I	Free Elective I (slot) - never taken	3 (0 here)	U-FE	Elective slot TBD	-	-	-	P (gap) / A (moved)	STR/AUD	Not in audit. Moved to SEM 7

Y1 S2 - 2024/2025 Sem II (202420252) - completed

Code	Course (English name)	Cr	Cat	Status	Grade / section	Prerequisite	Prereq met?	Evidence	Source	Note
SCSE1203	Software Engineering Principles	3	T	Passed (B)	B	SCSE1013	Yes (C)	P	AUD/STR	
SCSE1224	Advanced Programming	4	T	Passed (C+)	C+	SCSE1013	Yes (C)	P	AUD/STR	
SCSR1033	Computer Organization and Architecture	3	T	Passed (C+)	C+	SCSR1013	Yes (C+)	P	AUD/STR	
SCSR2213	Network Communications (1st attempt)	3 (0 here)	T	Failed - superseded by repeat	D (G)	-	-	P	AUD	Repeated 2025/26-II -> B-
SCST1223	Probability and Statistical Data Analysis (1st attempt)	3 (0 here)	T	Failed - superseded by repeat	D (G)	-	-	P	AUD	Repeated 2025/26-II -> C-
UHLM1012	Malay Language for Communication 2	2	U	Passed (A-)	A-	-	-	P	AUD	International-student version of ULRS1182

Y2 S1 - 2025/2026 Sem I (202520261) - completed

Code	Course (English name)	Cr	Cat	Status	Grade / section	Prerequisite	Prereq met?	Evidence	Source	Note
SCSE2103	Data Structure and Algorithm	3	T	Passed (B)	B	SCSE1013	Yes	P	AUD/STR	
SCSE2123	Software Requirements Engineering	3	T	Passed (C+)	C+	SCSE1203	Yes	P	AUD/STR	
SCSE2133	Software Process and Project Management	3	T	Passed (B-)	B-	SCSE1203	Yes	P	AUD/STR	
SCSM2113	Human Computer Interaction Fundamentals	3	T	Passed (B-)	B-	SCSE1203	Yes	P	AUD/STR	
SCSR2043	Operating Systems	3	T	Passed (A-)	A-	-	-	P	AUD	
ULRF2622	Football (Service Learning & Community Engagement slot ULRF2xx2)	2	K	Passed (A)	A	-	-	P	AUD/STR	

Y2 S2 - 2025/2026 Sem II (202520262) - completed (GPA 2.22, CGPA 2.55)										KD after: 57 - P
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE2243	Application Development Project I	3	T	Passed (A)	A	SCSE1203, SCSE2123	Yes	P	RES/AUD	
SCSM2223	Cross-Platform Application Development	3	T	Passed (B)	B	SCST1143	Yes	P	RES/AUD	
SCSR2213	Network Communications (repeat, UK)	3	T	Passed (B-)	B-	-	-	P	RES/AUD	
SCST1223	Probability and Statistical Data Analysis (repeat, UK)	3	T	Passed (C-)	C-	-	-	P	RES/AUD	
SCSE2233	Software Design and Architecture	3 (0 here)	T	Repeat needed	D (G, UM)	SCSE1203	Yes	P	RES/AUD	Not earned. Repeat planned in SEM 6
SCSP3213	Business Intelligence (Y2S2 programme elective)	3 (0 here)	E	Repeat needed	D (G, UM)	-	-	P	RES/AUD	Not earned. Repeat planned in SEM 6
UHLB2122	Professional Communication Skills 1 (Y2S2 in [STR])	2 (0 here)	B	Not yet taken	-	-	-	P	STR	Moved to SEM 5 as 'to add'
ULRS1022	Philosophy and Current Issues (Y2S2 in [STR])	2 (0 here)	U	Not yet taken	-	-	-	P	STR/AUD	Not in audit. Moved to SEM 7

SEM 5 - Y3 S1 - 2026/2027 Sem I (202620271) - CURRENT: 12 Oct 2026 - 21 Mar 2027										18 credits | KD after: 69 (registered only) / 75 (with adds) - C
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE3103	Cognitive Computing	3	E	Registered (slip)	sec 05	-	-	P	SLIP/AUD	
SCSE3143	Ubiquitous Computing	3	E	Registered (slip)	sec 05	-	-	P	SLIP/AUD	
SCSE3203	Special Topics	3	E	Registered (slip)	sec 15	-	-	P	SLIP/AUD	Time/room still unknown
SFMX1063	Information Technology in Business (fills Free Elective II slot)	3	U-FE	Registered (slip)	sec 02, FC, NFE	-	-	P	SLIP/AUD	Audit lists it under U (Umum Universiti). It IS the FE II slot - no extra FE II row
UHLB2122	Professional Communication Skills 1	2	B	To add (planned, unconfirmed)	add 15 Oct	-	-	A	MOH/OLD	Not on slip of 04 Oct; date not in official sources
UHLB3132	Professional Communication Skills (2)	2	B	To add (planned, unconfirmed)	add 15 Oct	none listed in [STR]	Yes (none listed)	A	MOH/OLD/ STR	Not on slip; check if UHLB2122 must come first
UHLF1112	French Language (French 1) - foreign language elective	2	B	To add (planned, unconfirmed)	sec 13 Wed 9-11am	-	-	A (status) / P (section)	FL/MOH	Late reg. by 14 Oct (not in official sources). See K1 before adding

SEM 6 - Y3 S2 - 2026/2027 Sem II: 22 Mar - 3 Oct 2027 (lectures 22 Mar - 4 Jul, exams 12-25 Jul)										17 credits | KD after: 92 - C
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE2233	Software Design and Architecture - REPEAT	3	T	Repeat needed	was D	SCSE1203	Yes (B)	P (prereq) / A (offered Sem II: [STR] Y2S2)	STR/AUD	Unblocks SCSE3213
SCSP3213	Business Intelligence - REPEAT	3	E	Repeat needed	was D	-	-	A (offered Sem II: [STR] Y2S2)	STR/AUD	Alternative for the Y2S2 elective: SCSB2103 Bioinformatics I or SCSP2753 Data Mining (would not replace the D - A)
SCSR3133	Secure Software Programming	3	T	Not yet taken	-	'SCSM 2213' per [STR]	Likely (A): code not in [CUR]; SCSM2113 B-/SCSM2223 B passed	A	STR/CUR	Conflict K3
SCSE3223	Application Development Project II	3	T	Not yet taken	-	SCSE2243	Yes (A)	P (prereq) / A (sem)	STR/RES	
SCSE3233	Professional Practice in Software Engineering	3	T	Not yet taken	-	SCSE1203	Yes (B)	P (prereq) / A (sem)	STR/AUD	
ULRS3032	Entrepreneurship and Innovation	2	U	Not yet taken	-	-	-	A (sem)	STR	

Short Semester 2026/2027: 26 Jul - 3 Oct 2027 - nothing planned (contingency only)
Use only if a repeat (SCSE2233 / SCSP3213) or another SEM 6 course is failed again. Course offerings in the short semester are not in the sources (A).

SEM 7 - Y4 S1 - 2027/2028 Sem I (approx. Oct 2027 - Mar 2028; dates A)										13 credits | KD after: 105 - C
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE3242	Software Engineering Project I	2	T	Not yet taken	-	80 credits + SCSE2243	Yes: KD 92 >= 80 at start (C) + SCSE2243 A (P)	C (gate) / A (offered in Sem I)	STR/RES	[STR] places it in Y3S2. Earliest KD>=80 semester. Fallback F1 if Sem II only
ULRS1022	Philosophy and Current Issues	2	U	Not yet taken	-	-	-	A (sem; [CUR] lists Y1 S1)	STR/CUR	
FE I	Free Elective I (slot)	3	U-FE	Elective slot TBD	-	non-Computing faculty	-	A	STR/CUR	Any course in [CUR] UNIVERSITY list offered by a faculty other than Computing ([STR] footnote)
FE III	Free Elective III (slot)	3	U-FE	Elective slot TBD	-	non-Computing faculty	-	A	STR/CUR	Any course in [CUR] UNIVERSITY list offered by a faculty other than Computing ([STR] footnote) ([STR] puts it in Y4S2)
FE IV	Free Elective IV (slot)	3	U-FE	Elective slot TBD	-	non-Computing faculty	-	A	STR/CUR	Any course in [CUR] UNIVERSITY list offered by a faculty other than Computing ([STR] footnote) ([STR] puts it in Y4S2)

SEM 8 - Y4 S2 - 2027/2028 Sem II (approx. Mar - Jul 2028; dates A)										13 credits | KD after: 118 - C
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE4214	Software Engineering Project II	4	T	Not yet taken	-	SCSE3242	Planned (SEM 7)	A	STR	
SCSE3213	Software Quality and Testing	3	T	Not yet taken	-	SCSE2123, SCSE2233	SCSE2123 yes (C+); SCSE2233 planned SEM 6	A	STR	Cannot be in SEM 6: SCSE2233 must be passed first
PE-A	Programme Elective (Y4S2 group) #1	3	E	Elective slot TBD	-	-	-	P (options) / A (choice)	STR	SCSR4453 Network Security; SCSR4973 Computer Network & Security Special Topics; SECB3133 Computational Biology I; SECB3203 Programming for Bioinformatics; PRISMS Elective (SECR/SECP/SECJ5xx3, eligibility per PRISMS handout)
PE-B	Programme Elective (Y4S2 group) #2	3	E	Elective slot TBD	-	-	-	P (options) / A (choice)	STR	SCSR4453 Network Security; SCSR4973 Computer Network & Security Special Topics; SECB3133 Computational Biology I; SECB3203 Programming for Bioinformatics; PRISMS Elective (SECR/SECP/SECJ5xx3, eligibility per PRISMS handout)

SEM 9 - 2028/2029 Sem I (approx. Oct 2028 - Feb 2029; dates A) - Industrial Training										12 credits | KD after: 130 - C
Code	**Course (English name)**	**Cr**	**Cat**	**Status**	**Grade / section**	**Prerequisite**	**Prereq met?**	**Evidence**	**Source**	**Note**
SCSE4108	Industrial Training (HW)	8	T	Not yet taken	-	92 credits + CGPA >= 2.0	Yes: KD 118 (C); CGPA 2.55 now (P) - must stay >= 2.0	C / A (sem)	STR	
SCSE4114	Industrial Training Report	4	T	Not yet taken	-	92 credits + CGPA >= 2.0	Yes (same as above)	C / A (sem)	STR	

'(0 here)' = the row is shown for history but its credits are counted in another semester (repeat or moved slot) or are outside the 128.

Assumptions (A) and unresolved conflicts (K)

ID	Assumption
A1	Offering semester = where [STR] places the course (Year x Sem 1 = Sem I in Oct, Sem 2 = Sem II in Mar). [STR] does not state offerings.
A2	SCSE3242 Software Engineering Project I is offered in Sem I (SEM 7). Fallback F1: if it runs only in Sem II, swap: SEM 7 = Industrial Training SCSE4108 + SCSE4114, SEM 8 = SCSE3242 + SCSE3213 + ULRS1022 + FE I/III/IV, and SCSE4214 + 2 programme electives in SEM 9 (if offered in Sem I) or SEM 10 (Mar-Jul 2029). SEM 10 matches the 29 Aug 2029 expected graduation.
A3	The three SEM 5 adds (UHLB2122, UHLB3132, UHLF1112) get registered. If not, KD after SEM 5 = 69 (not 75), after SEM 6 = 86: still >= 80 for SCSE3242 in SEM 7. The 6 B credits then need a later semester (offerings of UHLB/UHLx in other semesters not in sources).
A4	Every course from now on is passed at the first try. Each failure removes its credits from all later KD figures.
A5	A repeat replaces the old D in the CGPA (UM/UK). Calculated support: JMN 155.66 / KK 61 come out exactly when the first D attempts of SCSR2213 and SCST1223 are left out. The official rule is not in the sources.
A6	The '80 credits' (SCSE3242) and '92 credits' (Industrial Training) gates are measured as official KD (which includes UHLB1112). On the 128-basis the numbers are 2 lower (90 after SEM 6), but the gates still hold where they are used.
A7	Dates for 2027/2028 and 2028/2029 are estimated from the 2026/2027 pattern. Only the 2026/2027 calendar is in the sources.
A8	Which slot SFMX1063 fills (FE II) is a labelling choice. The audit only lists it under U (Umum Universiti); the credit effect is the same.
A9	Loads of 12-18 credits are allowed. The maximum credit load rule is not in the sources.

ID	Conflict / open question
K1	UHLB1112 English Communication Skills: the audit lists it under B (Bahasa Inggeris, 6) and counts its 2 credits in KD 57 (49 T + 4 U + 2 K + 2 = 57). [STR] treats it as an extra course for students with low English scores. This plan keeps B6 = UHLB2122 + UHLB3132 + UHLF1112 (plan total 128; official KD would end at 130). ASK THE ADVISOR TODAY: if UHLB1112 counts toward B6, only 4 more B credits are needed and one of the three adds (e.g. UHLF1112) may be unnecessary.
K2	[CUR] lists UKQA2022 Digital Media & Community (2) under FACULTY CORE, which makes 84 against the 82 heading. [STR] and the audit do not list it, so it is excluded here. Advisor to confirm.
K3	SCSR3133 Secure Software Programming prerequisite 'SCSM 2213' ([STR]) is not a course in [CUR]. It is probably SCSM2113 or SCSM2223, both passed.
K4	Graduation timing: [STR] = 8 semesters (Jul 2028). This plan ends after SEM 9 (approx. Feb 2029). Status letter (via Omar) says expected graduation 29 Aug 2029; fallback F1 also fits that date.
K5	PRISMS codes differ: [STR] SECR/SECP/SECJ5xx3 vs [CUR] SCSR/SCSP/SCSJ5xxx. PRISMS eligibility (likely CGPA-based) is not in the sources; at CGPA 2.55 it may not be open.
K6	[STR] names the foreign-language slot 'UHLx1122'; [CUR] and [FL] list UHLF1112 French Language under ENGLISH 6.
K7	[CUR] UNIVERSITY list includes SCSE4213 / SCSE4223 (Computing courses), but the [STR] footnote says free electives must come from other faculties. Pick FF courses from non-Computing faculties.
K8	Year label: slip says 'YEAR 2', audit says 'BIL SEMESTER 5'. No effect on credits.

Sources (paths relative to Library unless stated)

[AUD]	Academic-Records\Graduation-Audit.pdf (scanned, read visually; printed 04 Oct 2026)
[RES]	Academic-Records\2025-2026\Semester-2\2025-2026-Sem-2-Exam-Results.pdf (KD 57, KK 61, JMN 155.66, CGPA 2.55)
[SLIP]	Academic-Records\2026-2027\Semester-1\2026-2027-Sem-1-Course-Registration.pdf (04 Oct 2026, 12 credits)
[CCS]	Academic-Records\Course-Completion-Status.pdf
[STR]	Academic-Program\SCSEH-Course-Structure-Intake-2024-2025-Rev-2024-11-05.pdf
[CUR]	Academic-Program\SCSEH-Curriculum-Course-List.pdf
[CAL]	Academic-Calendar-and-Timetables\Academic-Calendar-2026-2027-Corrected-2026-05-14.pdf
[FL]	Academic-Calendar-and-Timetables\Foreign-Languages-Timetable-2026-2027-Sem-1.pdf
[OLD]	Academic-Program\Mohammed-SCSEH-Study-Plan-2026-10-12.pdf (old AI-made plan, not official)
[MOH]	Mohammed's own statements/corrections (planned adds, colour meaning) - not a document
[OMAR]	The_Boots\01_Omar\STATE\CURRENT.md + OUTPUT\Mohammed_SCSEH_Plan_review_2026-10-13.md (status letter info: expected graduation 29 Aug 2029)

Before the next deadlines: (1) Ask advisor Muhammad Luqman bin Mohd Shafie about K1 (does UHLB1112 count in B6?) before the UHLF1112 late registration (14 Oct) and the UHLB adds (15 Oct). (2) Confirm that SCSE3242 Software Engineering Project I runs in Sem I (A2); if not, use fallback F1. (3) Confirm the repeat offerings in 2026/27 Sem II.